PATRIZIO ACQUADRO

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Robot Learning & Manipulation: multimodal perception, simulation, and real-robot systems

Skills

Core: Python, PyTorch, Isaac Sim/Lab, MuJoCo, HF Transformers, C, Java, NumPy, Pandas, SQL, R.
Tools & Software: Git, Linux, Docker, Apptainer, Slurm, DeepSpeed, Weights & Biases, LangChain.
Languages: Italian (native), English (fluent; IELTS 7.5), French (basic; 2 years, 6 hours/week).

Work & Research Experience

R&D Assistant, Robot Learning – Purdue University (E-Lab) Jun 2026 – Present
- Build simulation and control infrastructure for humanoid manipulation, and perception systems supporting ONR research.
- Mentor an undergraduate researcher on Ego2Grip, guiding technical development.

Visiting Student Researcher – Purdue University (E-Lab) Jan 2026 – May 2026
- Integrated a decoder-only transformer policy on the IHMC Alex humanoid for real-time validation.
- Owned the end-to-end experimentation pipeline, optimizing for reproducibility, tracking, and rapid ablations.

Generative AI Researcher – Politecnico di Milano Feb 2025 – Sep 2025
- Developed RAG code generation with retrieval/reranking over 7.6k code snippets and 4-bit LLMs.
- Built a LangChain coding agent with 5 tools and co-designed a 15-task repository benchmark.
- Ported an INT8 transformer to a CNN-oriented NPU (STM32N6); validated with ST engineers.

Applied ML Researcher – University of Milano-Bicocca Mar 2024 – Jun 2024
- Designed LangChain RAG with pairwise re-ranking to improve clinical-trial ranking and reduce noise.
- Built a UMLS-based KB for grounding and benchmarked OSS vs GPT API on accuracy/latency/cost.

AI Research Intern (Quantum ML) – Exprivia S.p.A. Apr 2024 – Jun 2024
- Evaluated QML feasibility for Heston modeling by benchmarking PINN/WGAN baselines against a qWGAN in Qiskit.

Selected Projects & Entrepreneurship

AlexDoor-XAS: Compared four action representations for humanoid door manipulation across ACT, diffusion, and VLAs.

Isaac Audio Sensors: Developed an open-source SDK to enable robot audition in Isaac Sim/Lab.

SquadBot-AV: Built an audiovisual scene graph, temporal world model, and learned policies for the IHMC Alex humanoid.

Ego2Grip: Retargeting 3D hand motion from first-person video to Alex grippers for accessible manipulation demonstrations.

VLA-LEGO: Built a contact-rich MuJoCo benchmark and VLA architecture for humanoid bimanual LEGO assembly.

BiellaInFesta – Founder Jul 2022 – Present
Grew a local events page to 10k+ followers and 800k views/mo; lead a 4+ person team.

Adunet WebApp – Founder Apr 2025 – May 2025
Shipped a Next.js festival map app for 8–10k daily visitors; led a 10-person promo team.

Education

Purdue University – Visiting Exchange Student Aug 2025 – May 2026
Selected on merit (1 of 2) from a 674-student first-year MSc CS cohort (PoliMi).

Politecnico di Milano – M.S. Computer Science & Engineering (AI) Sep 2024 – Present
GPA 27/30; Honors (30L/30): LLMs (PhD course), SWE II.

University of Milano-Bicocca – M.S. AI for Science and Technology Sep 2024 – Present
Top grades (30/30): Advanced AI/ML/DL, Embedded Systems.

University of Pavia – B.S. Artificial Intelligence Sep 2021 – Oct 2024
Degree grade: 110/110; First inter-university B.S. in AI in Italy.

University Scholarships: Merit-based, income-qualified (awarded 2023–2024–2025, €5,500/year)

Additional

Interests: Swimming (3 years competitive), rock climbing (regional competitions), philosophy, and fantasy literature.
Community: JCI Biella: led communications for 4 events; Italian Red Cross: certified first-aid volunteer.